MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

1/ TYPE DOD-039

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Forward-wave amplifier, 2.9 to 3.1 GHz operation, 4 kw peak power output, cathode pulsed, periodic-permanent magnet focused.

ABSOLUTE RATINGS: See notes 1, 2, 3, and 4

Parameter:	Ef	If(surge)	ik	ek	Pd	Pp	Du	tp	tk	Alt (operating)
Unit:	V	A	a	kv	W	W	---	µs	sec	ft
Maximum:	6.6	10	2.1	-9.5	2	750	0.03	40	---	10,000
Minimum:	6.0	---	1.5	-8.5	---	---	0.007 (Note 5)	--- (Note 5)	180	---

ABSOLUTE RATINGS: See notes 1, 2, 3, and 4

Parameter:	Coolant flow rate	Coolant inlet temp Water	Coolant inlet temp EG & W	Coolant pressure	Coolant pressure drop	TA	F
Unit:	GPM	°C	°C	psig	psig	°C	GHz
Maximum:	0.5	62	14	200	15	-54	F3
Minimum:	0.45	---	---	---	10	+65	F1

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 4
DC connectors:	See figure 1	Construction:	Metal and ceramic
RF connectors:	See figure 1	Weight:	11 pounds, (5 kg) maximum
Cooling:	Liquid, see note 6	Mounting positions	Any
Focusing:	Periodic permanent magnet	Shelf life guarantee:	See note 7

TEST CONDITIONS: See notes 4 and 8

Parameter:	Ef	Pd	o'	Du	tp	tk	F
Unit:	V	W	---	---	µs	sec	GHz
Tolerance:	---	max	min	---	±1	min	---
	6.3	0.7	1.1:1	0.03	37	180	F2

Frequency	
F	GHz
1	2.9
2	3.0
3	3.1

GENERAL:

Qualification - Required

1/ Replaces Varian Type VAS - 805T7, ITT Gilfillan P/N's 145005-1, 145005-2, and tube type 8742.

(B) denotes changes

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position. The sum of the readings should equal 360 degrees or 6400 mils. Any

	Qualification inspection						
1011	Humidity	9	No voltages applied	---	---	---	---
1027	Temperature cycling (nonoperating)	10	TA = -54° to +75°C; 5 cycles	---	---	---	---
---	Vibration	11, 12		---	---	---	---
1042	Shock	12	Test condition A; 6 shocks (see figure 1)	---	---	---	---
4258	Radio interference shielding	13		---	---	1.0	mW/m^2
(B) 1136	Rough handling	9		---	---	---	---
	Quality conformance inspection, part 1	14					
(5) 4289	Heater current (nonoperating)	-	Ef = 6.3 V	If	2.0	4.0	A
(B) 1256	Electrode current (cathode)	15		ik	1.54	1.90	A
1261	Electrode voltage (cathode)	5		ek	-8.5	-9.5	kv
4214	Cathode emission	16	tk1 = 250; tk2 = 300	Δik	---	0.1	a
4253	Gain (large signal)	17	F1 thru F3	Gls	37.55	---	dB
4254	Saturated rf power output	17	F1 thru F3	po	4.0	---	kw
4253	Gain variation (large signal)	17, 18	F1 thru F3	ΔGls	---	1.0	dB
4261	Spurious output or stability	19		spo	---	-40	dB
---	Stability	20	F1 thru F3	---	---	---	---
4256	Input and output match (cold)	21	F1 thru F3	VSWR	---	2.2:1	---
4257	Amplifier insertion loss	-	F = F2	Li	50	---	dB
1155	Coolant pressure drop versus coolant flow	6	Inlet temp = 62°C (max); flow rate = 0.5 GPM	ΔP	10	15	psig
---	Coolant pressure	-		P	---	200	psig
	Quality conformance inspection, part 2						
4278	Phase shift	22	Δek = ±50v	Δθ/Δek	---	0.3	deg/v
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
(B) ---	Life test	7, 8	Group D	t	1,000	---	hrs
---	Life-test end point:						
4254	Saturated rf power output	3,17, 23		po	3.6	---	kw

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NOTES:

1. In addition to the MIL-E-1 symbols, the following shall apply:

EG & W - - - - - - - - - -	Ethylene glycol and water
GPM- - - - - - - - - - - -	Gallons per minute
P- - - - - - - - - - - - -	Pressure
ppm- - - - - - - - - - - -	Parts per million
psig - - - - - - - - - - -	Pounds per square inch, gauge
Δek- - - - - - - - - - - -	Change in cathode pulse voltage
ΔP - - - - - - - - - - - -	Pressure drop
Δθ - - - - - - - - - - - -	Change in phase

2. Cathode and one leg of heater shall be connected internally. The helix is operated at ground potential. All voltages are referred to ground except the heater potential which is referred to the cathode.

3. The tube shall satisfy all requirements of this tube specification sheet (TSS) when tested in a line type modulator (figure 2, or equivalent), employing a pulse transformer (ITT Gilfillan Part No. 160020, or equivalent). The line type test modulator shall utilize a pulse forming network whose impedance is within ±10 percent of that shown on figure 2.

4. A copy of the critical performance characteristics including operating voltages shall be shipped with each tube. The operating voltages shall also be clearly indicated on the tube label and on a small decal at the collector end of each tube. Each tube delivered on the contract shall operate optimally under a single combination of discrete voltages which fall within the ranges shown under absolute ratings. All operating requirements of this tube specification sheet shall be met under these specific voltages.

5. The pulse duration (tp) shall be defined as the time interval between the first and the last instants at which the instantaneous amplitude reaches 90 percent of the peak pulse amplitude. The combination transition period of rise and fall times of the pulse, between the 30 and 90 percent amplitude points, shall not exceed 5.0 microseconds with a duty cycle of 0.03.

6. Distilled water with a maximum inlet temperature of +62°C shall be used with, (1) specific resistivity maintained greater than 0.5 megohm, (2) oxygen content maintained less than 0.5 ppm, (3) an organic filter, and (4) a 3 to 4 micron filter at ambient temperatures in excess of +4°C. A 60 percent ethylene glycol and 40 percent distilled water solution, by volume, with a maximum inlet temperature of +14°C, with the same resistivity, oxygen content, and filters shall be used at ambient temperatures of +4°C and lower.

7. The tubes tested and procured in accordance with this tube specification sheet, shall be capable of shelf-life for a period of 12 months after date of shipment from the manufacturer's plant. Tubes placed in operation during this shelf-life period, shall meet all requirements of this tube specification sheet. Tubes found to be defective, within the shelf-life period, shall be replaced by the manufacturer at no cost to the Government, or its contractors.

8. The tube shall be operated in the saturated mode. Test condition shall be defined as operation with power input of not more than 0.7 watt peak and the cathode operated at the optimum voltage.

9. After the performance of the humidity and rough handling tests, the tube shall be capable of passing the following quality conformance inspection tests:

Method	Test
4253	Gain (large signal)
4261	Spurious output or stability
4278	Phase shift

10. At the conclusion of this test, the TUT shall be capable of satisfying the requirements of quality conformance inspection, parts 1, 2, and 3 tests.

11. The tube shall be rigidly mounted to the vibration table and shall be vibrated in accordance with MIL-STD-167(SHIPS), Type I, Environmental vibration, except the amplitudes shall be as indicated below. The detected output of the tube shall be displayed on an oscilloscope.

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NOTES: -Continued

11. -Continued

Prior to vibration, the oscilloscope shall be calibrated in the dc mode, using slow sweep, by means of a precision attenuator. The face of the display tube shall be marked with lines corresponding to the saturated power output level, and to levels of ±0.5 dB. The attenuator then shall be returned to the saturated power output level. The displayed variations attributable to vibration shall be within the calibration lines.

Frequency	Amplitudes
5 to 15 Hz	0.075 ± 0.012 (0.150 Total displacement)
16 to 25 Hz	0.050 ± 0.008 (0.100 Total displacement)
25 to 33 Hz	0.025 ± 0.004 (0.050 Total displacement)

12. The tube shall exhibit interruptions or discontinuities in power output and shall be cable of satisfying all quality conformance inspection tests without damage or failure before and after performance of the specified test.

13. The power density shall be measured over the frequency range 0.8 to 12.9 GHz at a distance of 1 meter from the tube in the three principal planes for both horizontal and vertical polarization. The input signal level shall be the minimum required to saturate the tube over the frequency range F1 to F3.

14. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

(B) 15. The cathode current shall be measured at the optimum cathode voltage (label value) with a pi(rf) of 0.7 watt maximum, and shall be such that the effective beam Z is within 4,987 to 5,513 ohms.

16. The change in cathode current (Δik) shall be the difference between ik1 and ik2 (corresponding to tk1 and tk2) and shall not exceed the maximum specified.

(A) 17. A swept-frequency technique may be used for this measurement at intervals not to exceed 20 MHz intervals.

18. The gain variation (large signal) shall be the difference between the maximum gain and the minimum gain at the specified test frequencies.

19. The input of the tube shall be terminated with a matched load, and the output shall be connected to a +10 dB directional coupler. The signal from the arm of the coupler shall pass through two filters connected in series (a high pass with a cutoff at 2.7 GHz and a low pass with a cutoff at 3.3 GHz), into a crystal detector. The output of the coupler shall be connected through a 3-dB attenuator to a variable-phase short circuit. The spurious power output shall be the maximum signal detected at the crystal as the phase of the short is varied through the one-wave length at 3 GHz.

20. The tube shall be stable when operating at the optimum voltage given on the test data sheet for the specific tube, and for all cathode currents within ±5 percent of that cathode current shown on the data sheet. In addition, with an input VSWR of 1.5 or less, the tube shall be capable of driving a mismatched load of 1.5:1.

21. The VSWR and frequency at both input and output shall be measured and recorded at the point of maximum VSWR in the f equency band. The tube shall operate with an input/output impedance of 50 ohms.

22. The cathode voltage shall be varied within the values specified. The $\Delta\theta/\Delta ek$ measurement is the difference between the phase shifts through the tube at the two cathode voltage extremes divided by the change in voltage.

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NOTES: -Continued

23. The end of life point shall be defined as that point in time when the power output is reduced to less than 3.6 kw peak anywhere between 2.9 and 3.1 GHz with the tube operated at the recommended cathode voltage, working into a matched load (50 ohms) having a VSWR of 1.1:1 or less, and driven with a fixed input power of 0.7 watt maximum.

Custodians:
Army - EL
Navy - EC

Review activity:
DSA - ES

User activities:
Navy - AS, OS, MC, CG

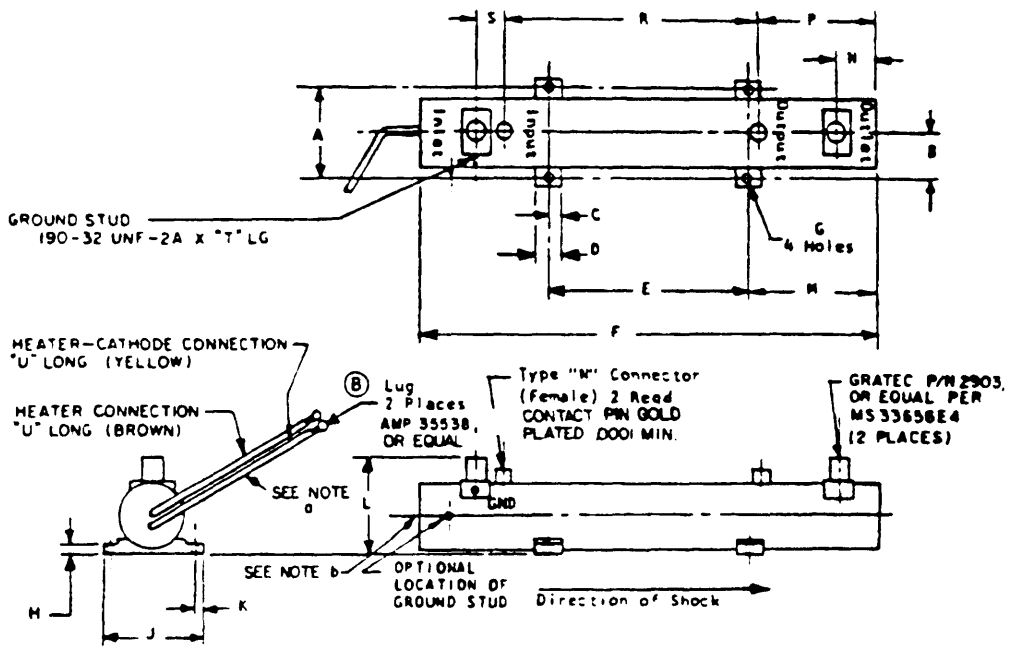
Preparing activity:
Navy - EC

Agent:
DSA - ES

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NOTES:

Ⓑa. Heater leads shall be Surpremant type SE-HV(30) 1927A(90) brown and yellow or ITT ETD P/N's 282359 (brown) and 282360 (yellow), or equivalent for use with terminal AMP 35538, or equal.

b. Heater leads omitted for clarity.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection		
A	3.365 (85.47)	3.385 (85.98)
B	1.630 (41.40)	1.750 (44.45)
C	.440 (11.18)	.560 (14.22)
D	.940 (23.88)	1.060 (26.92)
E	7.802(198.17)	7.822(198.68)
F	17.500(444.50)	18.000(457.20)
G	.271 (6.88)	.291 (7.39)
H	.350 (8.89)	.410 (10.41)
J	3.850 (97.79)	3.910 (99.31)
K	---	.250 (6.35)
L	---	4.250(107.95)
M	4.438(112.73)	5.500(139.70)
N	.812 (20.62)	1.500 (38.10)
P	2.312 (58.72)	3.750 (95.25)
R	10.438(265.13)	11.000(279.40)
S	1.250 (31.75)	1.875 (47.63)
T	.26 (6.60)	.50 (12.70)
U	19.70 (500.38)	20.30 (515.62)

FIGURE 1. Outline drawing of electron tube type DDD-

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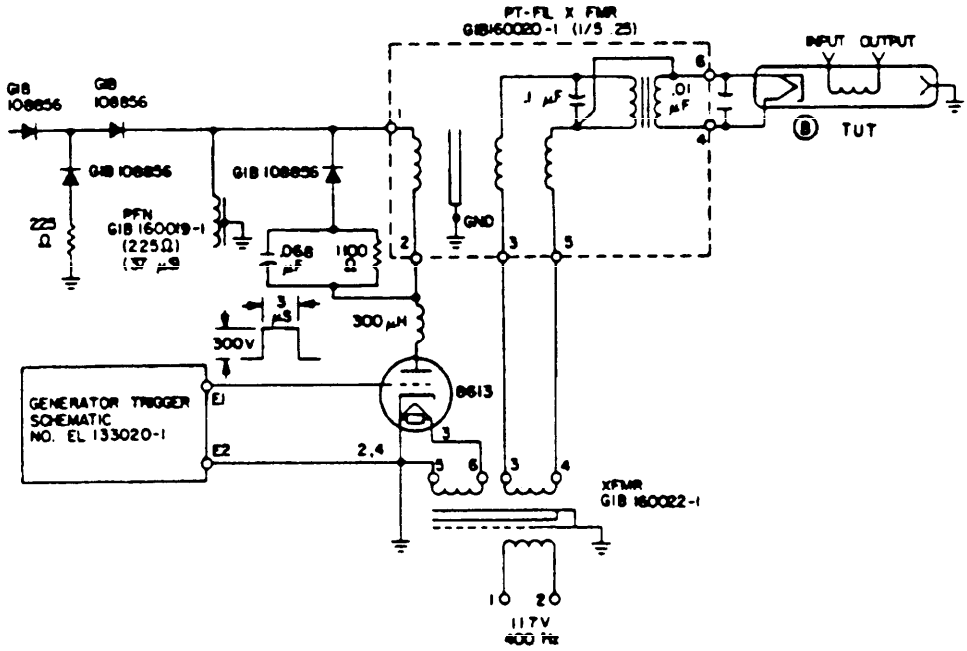


FIGURE 2. Test circuit for electron tube type 000-

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